



Fig. 4: Colour-coded dustbins for use in India

be a tiresome task. To tackle this manually, different countries have different colour-coded dustbins for dumping different waste materials. But manual separation of waste materials or throwing into separate trash bin is somewhat confusing as human beings can make mistakes. People can get confused as to where to dump the waste, because they are not sure of the actual material of the packaging and devices. This is where AI comes in.

Employing AI in garbage sorting and disposal processes is a better method for smart recycling and waste management. Therefore AI and various sensors including RFID tags are used. Many intelligent dustbins have been developed that are equipped with AI programs and Internet of Things (IoT) sensors in the waste management sector. Multitasking sorting AI robots have been developed for waste management that can sort tons of garbage on a daily basis.

In South Korean, RFID tags and sensors are used to collect garbage. A pneumatic garbage disposal system then reads these tags. The main computer, which stores all data, determines an appropriate method to dispose of the waste.

Chinese company Baidu and the United Nations Development Program are working to streamline the recycling of e-waste in China with a new app that can help users sell their old electronics for cash.

Rubicon Global has a cloud-based, full-service waste management and recycling solution. AI and machine learning are incorporated into the solution to enhance human insight and improve waste diversion. The company has also developed an app that uses computer vision to gener-

ate a quick assessment of the waste stream, and a device is mounted on garbage trucks to visually analyse roadside dustbins.

As per a report from *mirror.co.uk*, Bin.E is the world's first intelligent bin

that sorts waste materials. It uses a combination of sensors, image recognition and AI, and can recognise objects like glass, plastic or paper, and compress these before placing them in containers.

Developed by CleanRobotics, TrashBot robot is the first ever smart trash robot that uses AI to sort recyclables from landfill waste.

CP Manufacturing has a complete line of recycling equipment to ensure the highest recovery rates for virtually any material type, including commercial and industrial, e-waste, food waste and green waste recovery. L-VIS is a high-resolution colour camera sorting system from CP Manufacturing that provides accurate colour and size separation of small particles, and is suitable for electronic scrap, plastic flake, metal scrap, glass and other small particle segregation.

Finnish company ZenRobotics has come up with the world's first robotic waste sorting system. It employs AI for smart recycling of waste materials. Using computer vision, machine learning and AI, the robot can sort and pick recycled materials from moving conveyor belts.

e-Waste management in India

The government of India has issued revised E-Waste Management Rules 2016, which came into force in October 2016. The Pollution Control Board of India has suggested different colours for different dustbins, namely, green for biodegradable waste, black for e-waste, and blue for plastic and metals. But these dustbins are not used properly due to either ignorance or lack of seriousness of people. Black dustbins for dumping e-waste are rarely found on the streets. So, until now, most of the garbage and e-waste from Indian cities are in landfills or

have gone down the rabbit hole.

Absence of strict regulations on e-waste management, lack of awareness by the masses and absence of efficient infrastructure are some factors for the widespread e-waste. Lack of proper procedures for the collection, disposal and recycling are the main reasons for the poor state of e-waste management in India.

India still lags behind other countries when it comes to an automated waste management system. Recently, many Indian companies have started working on e-waste management and recycling solutions. These include Namo e-Waste, Revive Electronic Waste, Attero Electronics, Let's Recycle, Karo Sambhav, Cerebra and E-Parisaara.

India ranks third in the world in terms of high-quality research publications in AI. However, currently, implementations of AI and machine learning in e-waste management on a large scale are hardly seen here. Current e-waste management systems are unable to efficiently deal with the massive amounts of e-waste generated every day. Automating the processes of sorting and disposal using AI and machine learning would help in handling e-waste more effectively. It is time to switch to AI and machine learning for smart recycling and e-waste management in India.

To sum up

Our lack of initiative in e-waste management has resulted in many recyclable precious items being thrown into dustbins, which have ended up in landfills. It is the need of the hour to reduce, reuse and recycle e-waste. Gold, silver and other precious metals can be obtained by recycling e-waste. Non-recyclable items can either be sent to landfills or burned (incinerated) under controlled environments.

We can expect a significant decrease in e-waste if traditional techniques are replaced with automated intelligent solutions and sorting AI-robots. As sensible individuals, we should reduce e-waste and make others aware of its value, in addition to the harmful effects of e-waste. This is important to save our lives and the environment. **EFY**